SECTION 6

ELECTROMAGNETISM

Chapter 18

Electrostatics

Force between charges

Coulomb's Law of Force Between Electric Charges

The electrostatic force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of their distance apart.

Mathematically

$$F \propto \frac{q_1 q_2}{d^2}$$

$$\text{or } F = \frac{k q_1 q_2}{d^2}$$

where k is a constant dependent on the medium between the charges in N m^2 C^{-2}, which is $9{\cdot}00 \times 10^9$ N m^2 C^{-2} for a vacuum
F is the force in N
q is the charge in C
d is the distance apart of the charges in m.

For more advanced work in electrostatics, k is often written as $\frac{1}{4\pi\varepsilon_o}$, where ε_o is called the permittivity of a vacuum which is 8.84×10^{-12} C^2 N^{-1} m^{-2}. Hence, Coulomb's Law can be written

$$F = \frac{1}{4\pi\varepsilon_o} \frac{q_1 q_2}{d^2} \text{ for charges in a vacuum.}$$

The following notes are important.

(a) The permittivity of air may be taken as that for a vacuum.
(b) If another medium exists between the charges, ε_o has to be replaced by ε, the permittivity of the medium.
(c) The charge on an electron is $1{\cdot}60 \times 10^{-19}$ C.
(d) The coulomb is a large unit of charge and more convenient values are microcoulombs ($1\mu C = 10^{-6}$ C) and picocoulombs ($1 \text{ pC} = 10^{-12}$ C).

Example 1

Find the electrostatic force between the proton and the electron in a hydrogen atom, assuming that the average radius of the atom is $5{\cdot}30 \times 10^{-11}$ m.

$d = 5{\cdot}30 \times 10^{-11}$ m
$k = 9{\cdot}00 \times 10^9$ N m^2 C^{-2}
$q_e = q_p = 1{\cdot}60 \times 10^{-19}$ C
$F = ?$

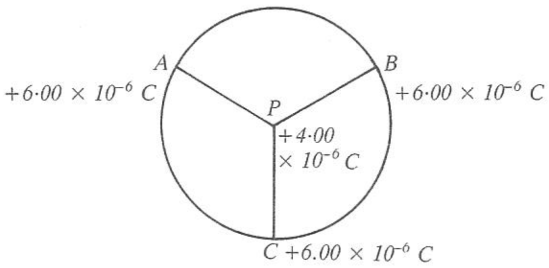
$$F = \frac{k q_e q_p}{d^2} = \frac{9{\cdot}00 \times 10^9 \times (1{\cdot}60 \times 10^{-19})^2}{(5{\cdot}3 \times 10^{-11})^2} = 8{\cdot}20 \times 10^{-8} \text{ N}$$

The charges are opposite in sign so the force of $8{\cdot}20 \times 10^{-8}$ N is attractive.

Example 2

Three identical +6·00 μC charges are situated at points on the circumference of a circle of radius 10·0 cm. The points are 120° apart. What is the magnitude of the force between any of these charges and a +4·00 μC charge at the centre of the circle, and what is the net force on the +4·00 μC charge?

$q_A = +6{\cdot}00 \times 10^{-6}$ C
$q_B = +6{\cdot}00 \times 10^{-6}$ C
$q_C = +6{\cdot}00 \times 10^{-6}$ C
$q_P = +4{\cdot}00 \times 10^{-6}$ C
$r = 0{\cdot}100$ m



$$F_{AP} = F_{BP} = F_{CP} = \frac{kq_1q_2}{d^2} = \frac{9{\cdot}00 \times 10^9 \times 6{\cdot}00 \times 10^{-6} \times 4{\cdot}00 \times 10^{-6}}{(1{\cdot}00 \times 10^{-1})^2} = 2{\cdot}16 \times 10^1 \text{ N}$$

Each of these forces will be directed outwards from P. To find the net force at P, find the vector sum of these forces. Since they are equal in magnitude and the angles between them are equal (i.e. 120°), their sum is zero. Hence $F_{net} = 0$.

Set 33

1 Find the electrostatic force between charges of +2·0 C and +5·0 C separated by a distance of 75 m in a vacuum.

2 Two charges of +8·0 mC and −6·0 mC attract each other with a force of $3{\cdot}0 \times 10^3$ N in a vacuum. What is the distance between the charges?

3 What charge will repel a charge of $+6{\cdot}4 \times 10^{-6}$ C with a force of $2{\cdot}7 \times 10^{-1}$ N when they are separated by 0·84 m in air?

4 Two identical charges of 1·20 μC are immersed in an oil, the distance between them being 68·4 cm. What is the permittivity of the oil if the force between the charges has a magnitude of 1·86 N?

5 Two small spheres, 25·0 cm apart in paraffin oil, carry charges of $+7{\cdot}00 \times 10^{-9}$ C and $+9{\cdot}00 \times 10^{-9}$ C. Find the force on each sphere if the permittivity of the paraffin oil is $4{\cdot}18 \times 10^{-11}$ C^2 N^{-1} m^{-2}.

6 What happens to the force between two charged metal spheres in a vacuum if the charge on each is doubled and the distance between them is multiplied by three?

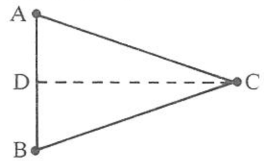
7 Two metal-coated table-tennis balls are charged identically and separated by a distance d in air. If the charge on one ball is doubled, the charge on the other tripled and reversed in sign, and the distance between them increased by 25%, how does the new electrostatic force compare to the old?

8 A metal sphere of mass $6{\cdot}0 \times 10^{-3}$ kg is found to just float in air above a similar metal sphere when both have a charge of 4·0 μC. Assuming that the only upwards force is electrostatic repulsion, find the distance between the spheres.

9 How many excess electrons must be placed on each of two small metal spheres placed 3·00 cm apart in a vacuum if the force of repulsion between the spheres is to be $8{\cdot}0 \times 10^{-19}$ N?

10 Two small metal spheres carry charges of +8·00 μC and −12·00 μC. The spheres are touched together and then separated to a distance of 7·50 cm in air. Determine the force between them.

11 Three charged spheres are fixed in positions as shown in the diagram.



The charge on each sphere is +6·0 nC.
AD = DB = 5·0 cm
DC = 14 cm

Find the net electrostatic force at C if the charges are in air.

12 Two very small metal-coated foam spheres, each of mass $2{\cdot}80 \times 10^{-6}$ kg, are attached to nylon threads 45·0 cm long and hung from a common point. When the spheres are given equal quantities of negative charge, each supporting thread makes an angle of 15·0° with the vertical. Find the charge on each sphere.

Answers

1 *$1{\cdot}6 \times 10^7$ N repulsion*
2 *12 m*
3 *$+3{\cdot}3 \times 10^{-6}$ C*
4 *$1{\cdot}32 \times 10^{-13}$ C^2 N^{-1} m^{-2}*
5 *$1{\cdot}92 \times 10^{-6}$ N repulsion*
6 *$\frac{4}{9}$ of original value.*
7 *3·8 times old force in the opposite direction*
8 *1·6 m*
9 *$1{\cdot}8 \times 10^3$*
10 *6·40 N repulsion*
11 *$2{\cdot}8 \times 10^{-5}$ N along DC*
12 *$6{\cdot}66 \times 10^{-9}$ C*

Electric fields

An electric field is a region in which an electric charge experiences a force due to its charge only. Electric fields are represented by lines of force, or flux lines. The strength of an electric field or the electric intensity at a point in the field is defined as the force per unit charge at that point and has the direction of the force on a positive charge at that point.

$$E_P = \frac{F}{q}$$

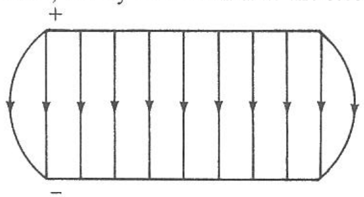
where E is the electric field strength at P in N C^{-1}
F is the electrostatic force in N
q is a unit positive charge at P in C.

Electric field is a vector quantity. The strength of the electric field at a distance d from a charge +q is given by

$+q \longleftarrow d \longrightarrow P\ (+q_1)$ $\quad E = \frac{F}{q_1} = \frac{kqq_1}{q_1d^2} = \frac{kq}{d^2}$ outwards

The electric field between oppositely charged plates is uniform. This is an important electric field and has useful applications. If the voltage between the plates is V and the distance between them d, it may be shown that the strength of the uniform field is given by

$$E = \frac{V}{d}$$



Hence, a N C^{-1} is equivalent to a V m^{-1}.

If a charged particle is introduced into a uniform field it will be accelerated uniformly by the field, gaining kinetic energy according to

$$\begin{aligned}\triangle E_k &= \tfrac{1}{2}mv^2 = \text{work done by field}\\ &= Fd\\ &= Eqd\\ &= \frac{V}{d}qd\\ &= Vq\end{aligned}$$

Since charged particles gain or lose kinetic energy in an electric field, an alternative unit of energy can be defined. This is the electron volt (eV). It is defined as the energy gained by one electronic charge when accelerated by a potential of one volt. If an electron is accelerated through 400 V in a TV tube, the energy gained is 400 eV. Note that 1 eV = $1{\cdot}60 \times 10^{-19}$ J.

Example 3

What is the strength of the electric field at a point 1·80 m from a charge of +3·00 μC?

$k = 9{\cdot}00 \times 10^9\ N\ m^2\ C^{-2}$
$d = 1{\cdot}80\ m$
$q = +3{\cdot}00 \times 10^{-6}\ C$
$E = ?$

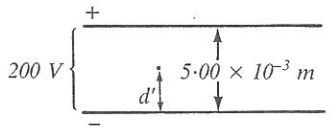
$$\begin{aligned}E &= \frac{kq}{d^2}\\ &= \frac{9 \times 10^9 \times 3{\cdot}00 \times 10^{-6}}{(1{\cdot}80)^2}\\ &= 8{\cdot}33 \times 10^3\ N\ C^{-1}\end{aligned}$$

The force is $8{\cdot}33 \times 10^3$ N C^{-1} outwards along the line joining the charge to the point.

Example 4

Two parallel metal plates are separated by a distance of 5·00 mm. If the voltage between the plates is $2{\cdot}00 \times 10^2$ V and the top plate is positive find:
(a) *the force on a charge of +1·50 μC placed midway between the plates, and*
(b) *the kinetic energy gained by the charge when it moves to the negative plate.*

$d = 5{\cdot}00 \times 10^{-3}\ m$
$V = 2{\cdot}00 \times 10^2\ V$
$q = +1{\cdot}50 \times 10^{-6}\ C$
$F = ?$
$\triangle E_k = ?$
$d' = \frac{5{\cdot}00 \times 10^{-3}}{2}\ m$



$$\begin{aligned}(a)\quad F &= Eq = \frac{Vq}{d}\\ &= \frac{2{\cdot}00 \times 10^2 \times 1{\cdot}50 \times 10^{-6}}{5{\cdot}00 \times 10^{-3}}\\ &= 6{\cdot}00 \times 10^{-2}\ N\end{aligned}$$

The force is $6{\cdot}00 \times 10^{-2}$ N towards the negative plate.

$$\begin{aligned}(b)\quad \triangle E_k &= Fd'\\ &= 6{\cdot}00 \times 10^{-2} \times 2{\cdot}50 \times 10^{-3}\\ &= 1{\cdot}50 \times 10^{-4}\ J\end{aligned}$$

The change in kinetic energy is $1{\cdot}50 \times 10^{-4}$ J.

Set 34

1 A charge of 12 C when placed in an electric field experiences a force of 648 N. What is the magnitude of the electric field strength?

2 A charge of 8·4 μC is introduced at a point in an electric field of intensity $1{\cdot}2 \times 10^2$ N C^{-1}. What is the magnitude of the instantaneous force on the charge?

3 An oil drop experiences an electrostatic force of magnitude $4{\cdot}48 \times 10^{-14}$ N when placed in a uniform electric field of strength $4{\cdot}00 \times 10^4$ N C^{-1}. What is the magnitude of the charge on the oil drop?

4 What is the electric intensity at a point 2·4 m from a point charge of 5·7 mC in air?

5 The strength of the electric field at a distance of 6·0 m from a small charged sphere is $3{\cdot}9 \times 10^5$ N C^{-1} towards the sphere. What is the charge on the sphere, assuming it is in air?

6 Two metal plates are oppositely charged and are separated by a distance of 4·5 mm in a vacuum. If a voltage of 168 V is connected across the plates, top plate positive, what is the strength of the electric field between the plates?

7 In a simulated Millikan oil drop experiment a charged foam sphere of mass $95{\cdot}2\ \mu g$ is found to float between two parallel plates separated by $1{\cdot}00$ cm in air and having 287 V across them. What is the magnitude of the charge on the foam sphere?

8 An electron in an electron gun of a TV picture tube is accelerated by a potential of $2{\cdot}5 \times 10^3$ V. Find:
(a) the kinetic energy gained by the electron in electron volts and joules
(b) the final speed of the electron, assuming it was initially at rest.

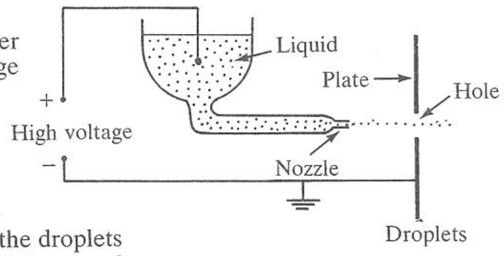
9 A double positively charged helium ion is accelerated to a speed of $2{\cdot}0 \times 10^7$ m s^{-1} in a linear accelerator by an electric field of magnitude $4{\cdot}5 \times 10^6$ V m^{-1}. How long will it take the ion to reach this speed from rest? (Mass of helium ion $= 6{\cdot}7 \times 10^{-27}$ kg)

10 Find the electric intensity at a point midway between charges of $+7{\cdot}2 \times 10^{-6}$ C and $-3{\cdot}4 \times 10^{-6}$ C which are $2{\cdot}0$ m apart in a vacuum.

11 Find the point between charges of $+8{\cdot}00\ \mu$C and $+6{\cdot}00\ \mu$C, which are situated $4{\cdot}00$ m apart in a vacuum, where the electric field strength is zero.

12 Charges of $+2.0\mu$C, $-4{\cdot}0\ \mu$C, $+4{\cdot}0\ \mu$C and $-2{\cdot}0\ \mu$C are placed at the vertices A, B, C, D of a square in the order given. If the square has a side of $1{\cdot}0 \times 10^1$ cm, what is the strength of the electric field at C?

13 In a device called a colloid thruster an intense electric field is used to atomise a non-conducting (dielectric) liquid into small droplets which are accelerated to a high velocity, producing enough thrust to manoeuvre a communication satellite.



If in such a colloid thruster the droplets, having a charge to mass ratio of $1{\cdot}6 \times 10^4$ C kg^{-1}, are accelerated to a speed of 36 km s^{-1}, find:
(a) the high voltage used
(b) the kinetic energy of the droplets if their diameters are $1{\cdot}0\ \mu$m and their density is $1{\cdot}4 \times 10^3$ kg m^{-3}.

14 An electron is accelerated from rest in the electron gun of a cathode ray oscilloscope by a potential of $1{\cdot}80 \times 10^3$ V. The electron then enters the region between two parallel and horizontal deflecting plates, which are $4{\cdot}00$ cm apart, and across which is applied a potential of $420{\cdot}0$ V. The electric field between the plates deflects the electron from its straight line path down the axis of the tube into a parabolic path.
(a) What is the horizontal velocity of the electron when it enters the deflecting plates?
(b) What is the horizontal velocity of the electron when it leaves the deflecting plates?
(c) How long does the electron spend travelling between the plates if they are $6{\cdot}60$ cm long?

(d) If the electric field accelerates the electron vertically downwards, what is its vertical displacement as a result of travelling between the plates?

Answers

1 54 N C^{-1} *2* $1{\cdot}0 \times 10^{-3}$ N *3* $1{\cdot}12 \times 10^{-18}$ C
4 $8{\cdot}9 \times 10^6$ N C^{-1} *away* *5* $1{\cdot}6 \times 10^{-3}$ C
6 $3{\cdot}7 \times 10^4$ V m^{-1} *down* *7* $3{\cdot}25 \times 10^{-11}$ C
8 (a) $2{\cdot}5 \times 10^3$ eV, $4{\cdot}0 \times 10^{-16}$ J (b) $3{\cdot}0 \times 10^7$ m s^{-1}
9 $9{\cdot}3 \times 10^{-8}$ s *10* $9{\cdot}5 \times 10^4$ N C^{-1} *towards negative charge*
11 $1{\cdot}86$ m *from* $6{\cdot}0\ \mu$C *charge* *12* $2{\cdot}8 \times 10^6$ N C^{-1} *at 17° left of CB*
13 (a) $4{\cdot}0 \times 10^4$ V (b) $4{\cdot}8 \times 10^{-7}$ J
14 (a) $2{\cdot}51 \times 10^7$ m s^{-1} (b) $2{\cdot}51 \times 10^7$ m s^{-1}
(c) $2{\cdot}63 \times 10^{-9}$ s (d) $6{\cdot}38 \times 10^{-3}$ m